How much of a shape does a person have?

The shape score, step by step

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Do I need to understand this?

You need the *idea*, not the algebra. The idea is three sentences:

1. The score is the amount of a shape that, added to the average person, matches the person as closely as possible.
2. It is found by multiplying the person's deviation by the shape at every position and adding the products up.
3. The larger the score (plus or minus), the more of the person's deviation that shape explains.

That is enough to read the figures, the model weights and the results. The algebra in Section 4 shows *why* the three sentences hold. It is useful if a supervisor asks, but in the thesis it is enough to say that the scores are functional principal component scores and cite the method.

1 The ingredients

Take one segment and one person. Along the segment, at positions s from 0 (start) to 1 (end):

- $d(s)$, the **deviation**: the person's turning angle minus the average person's turning angle at position s. Positive means the person turns more than average there.
- $\varphi(s)$, the **shape**: one fixed curve, found once from everybody. It is scaled so that $\sum_s \varphi(s)^2 = 1$; this only fixes its size and has no other meaning.
- c, the **score**: one number for this person. It is what we want to find.

To keep it small, use only three positions: start, middle and end.

position	start	middle	end
deviation d	3	1	-2
shape φ	0.6	0	-0.8

The person turns more than average at the start and less at the end. The shape says "more at the start, less at the end" (it crosses zero in the middle). Check the size: $0.6^2 + 0^2 + 0.8^2 = 1$. (Units of d are left out: think of 3 as 0.003 rad/mm.)

2 Trying amounts of the shape

We look for the amount c such that $c \times \varphi$ is as close as possible to d. For a given c, the miss at each position is $d - c\,\varphi$. We square the misses (so misses above and below count the same) and add them:

$$\text{error}(c) = \sum_s \big(d(s) - c\,\varphi(s)\big)^2.$$

Try $c = 2$: $c\,\varphi = (1.2,\ 0,\ -1.6)$, misses $= (1.8,\ 1,\ -0.4)$, error $= 3.24 + 1 + 0.16 = 4.4$.

Doing this for several amounts:

amount c	0	1	2	3	**3.4**	4	5
error	14	8.2	4.4	2.6	**2.44**	2.8	5

The error first falls, reaches its lowest value at $c = 3.4$, and rises again: too little of the shape leaves the deviation unexplained, too much overshoots. This is the U-shaped curve in column 4 of the week 10 figures. **The score is the bottom of that U: here $c = 3.4$.**

3 The shortcut: multiply and add

Instead of trying many amounts, multiply d and φ at each position and add:

$$c = \sum_s d(s)\,\varphi(s) = 3 \times 0.6 \ + \ 1 \times 0 \ + \ (-2) \times (-0.8) = 1.8 + 0 + 1.6 = 3.4.$$

The same number as the bottom of the U. Note what each position did:

- start: d and φ both positive, the product is positive and *adds*;
- middle: $\varphi = 0$, the position *does not count*;
- end: d and φ both negative, the product is positive and also *adds*.

A position where d and φ had opposite signs would have subtracted.

4 Why the shortcut gives the bottom of the U

Write out the square:

$$\text{error}(c) = \sum_s d^2 \ - \ 2c \sum_s d\,\varphi \ + \ c^2 \sum_s \varphi^2.$$

The shape has size 1, $\sum_s \varphi^2 = 1$, so

$$\text{error}(c) = \underbrace{\textstyle\sum_s d^2}_{14} \ - 2c\ \underbrace{\textstyle\sum_s d\,\varphi}_{3.4} \ + c^2.$$

This is a parabola in c. Its lowest point is where its slope is zero:

$$-2 \times 3.4 + 2c = 0 \quad \Longrightarrow \quad c = 3.4 = \sum_s d\,\varphi.$$

5 How much of the person the shape explains

Put the score back in:

$$\text{smallest error} = \sum_s d^2 - c^2 = 14 - 3.4^2 = 14 - 11.56 = 2.44.$$

The person's total deviation is $\sum_s d^2 = 14$. The shape removes $c^2 = 11.56$ of it, so it explains

$$\frac{c^2}{\sum_s d^2} = \frac{11.56}{14} = 83\,\% \text{ of this person's deviation.}$$

What is left (2.44, or 17 %) is column 5 of the figures.

6 More than one shape

Add two more shapes, each of size 1 and perpendicular to the others (the products of any two, added over the positions, are zero):

	start	middle	end	score $c = \sum d\,\varphi$
deviation d	3	1	-2	
φ_1	0.6	0	-0.8	3.4
φ_2	0.8	0	0.6	$2.4 - 1.2 = 1.2$
φ_3	0	1	0	1

Each score is found on its own, with the same multiply-and-add. Their squares add up to the whole deviation:

$$3.4^2 + 1.2^2 + 1^2 = 11.56 + 1.44 + 1 = 14 = \sum_s d^2.$$

So the three shapes explain 83 %, 10 % and 7 % of this person, and together all of it. With fewer shapes than positions (as in practice: 3 shapes, 50 positions) some deviation is always left over.

And the scores rebuild the deviation: $3.4\,\varphi_1 + 1.2\,\varphi_2 + 1\,\varphi_3 = (2.04 + 0.96,\ 1,\ -2.72 + 0.72) = (3,\ 1,\ -2) = d$.

7 What goes into the risk model

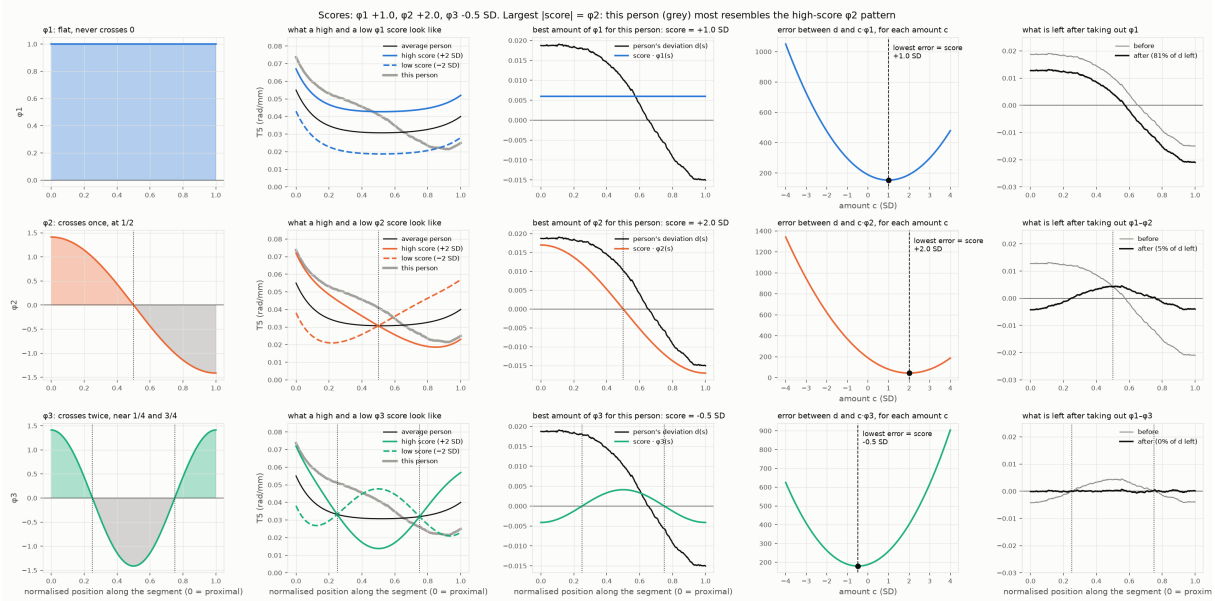
Only the scores. With three shapes each person enters the model as three numbers, and the model learns one weight per shape:

$$\text{cloglog}(p) = a + g\,\text{SCORE2} + b_1 c_1 + b_2 c_2 + b_3 c_3.$$

$b_k > 0$ means that people with more of shape k developed plaque more often.

8 On the real figures

The week 10 figure below does exactly this with 400 positions instead of three. Column 3 shows d and $c\,\varphi$, column 4 the error for each amount (the U), and column 5 what is left after removing each shape.


Scores: φ1 +1.0, φ2 +2.0, φ3 -0.5 SD. Largest |score| = φ2: this person (grey) most resembles the high-score φ2 pattern
φ1: flat, never crosses 0
what a high and a low φ1 score look like
best amount of φ1 for this person: score = +1.0 SD
error between d and c·φ1, for each amount c
what is left after taking out φ1
φ2: crosses once, at 1/2
what a high and a low φ2 score look like
best amount of φ2 for this person: score = +2.0 SD
error between d and c·φ2, for each amount c
what is left after taking out φ1–φ2
φ3: crosses twice, near 1/4 and 3/4
what a high and a low φ3 score look like
best amount of φ3 for this person: score = -0.5 SD
error between d and c·φ3, for each amount c
what is left after taking out φ1–φ3
average person
high score (+2 SD)
low score (−2 SD)
this person
person's deviation d(s)
score · φ1(s)
score · φ2(s)
score · φ3(s)
lowest error = score +1.0 SD
lowest error = score +2.0 SD
lowest error = score -0.5 SD
before
after (81% of d left)
after (5% of d left)
after (0% of d left)
T5 (rad/mm)
amount c (SD)
normalised position along the segment (0 = proximal)